

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT **Paper -2(Adv-2020-P2-Model)** **Date: 12-10-2025**

Time: 02.00Pm to 05.00Pm **GTA-03** **Max. Marks: 198**

12-10-2025_Sr.Super60_NUCLEUS-BT_Jee-Adv(2020-P2)_GTA-03_Syllabus,

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2020-P2-Model

Time: 3:00Hour's **IMPORTANT INSTRUCTIONS** **Max Marks: 198**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions With Integer Answer Type $	3	-1	6	18
Sec – II(Q.N : 7 – 12)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 25 – 30)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 43 – 48)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

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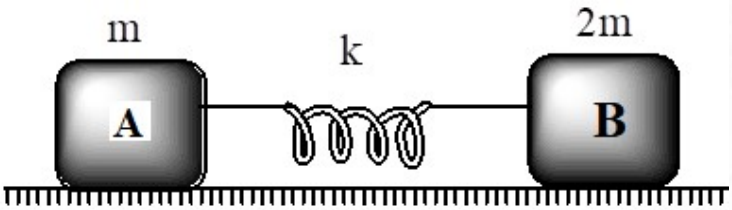
PHYSICS **Max Marks: 66**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

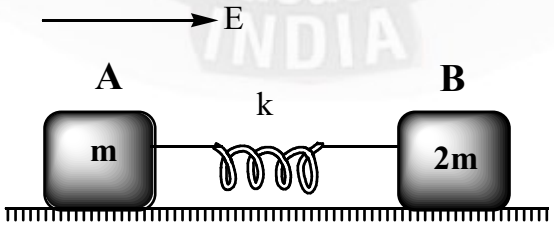
Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

1. A spring of spring constant 'k' is attached to two blocks A and B of masses $'m'$ and $'2\ m'$. The spring is initially compressed by x_o and released. Neglect friction at all contacts. work done by the spring on the block $'B'$ by the time the spring becomes undeformed is $\frac{1}{n}kx_0^2$, where n=



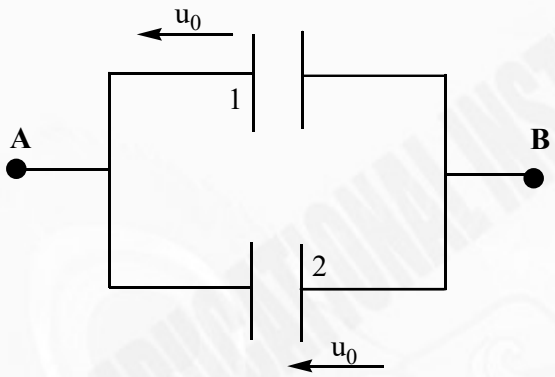
2. A uniform rod of length 2a is placed horizontally on a fixed thin horizontal rail at right angles to the rail and is released from rest. Initially the centre of the rod was at a distance a/3 from rail. If the rod slips after it has turned through an angle $\theta = 45^0$ then coefficient of friction μ between the rod and the rail is

3. In the figure shown two nonconducting blocks of A and B of mass $'m'$, $2m$ and charges $-Q$, $2Q$ respectively are attached at the two ends of a light spring of spring constant $'k'$. They are kept on a smooth horizontal surface. A and B are initially at rest and the spring is unstretched. Now a uniform electric field of intensity $'E'$ is switched on pointing towards right. Neglecting the electrostatic interaction between A and B, acceleration of $'B'$ at the moment of maximum extension in the spring is $\frac{QE}{z\,m}$ where z=

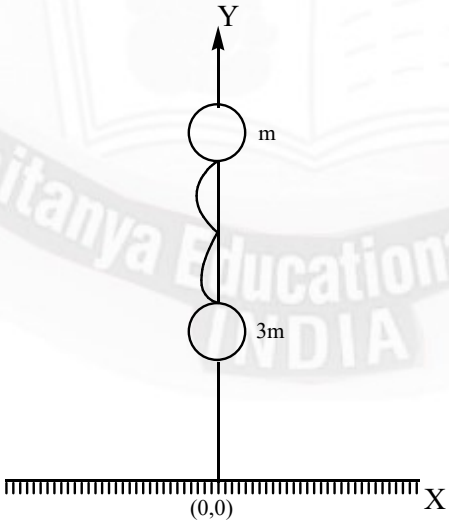


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4. Two identical capacitor having plate separation d_0 are connected parallel to each other across points A and as shown in figure. A charge Q is imparted to the system by connecting a battery across A and B and battery is removed. Now first plate of first capacitor and second plate of second capacitor starts moving with constant velocity u_0 towards left. Magnitude of current flowing in the loop during this process is $\frac{Qu_0}{x\, d_0}$, where x=

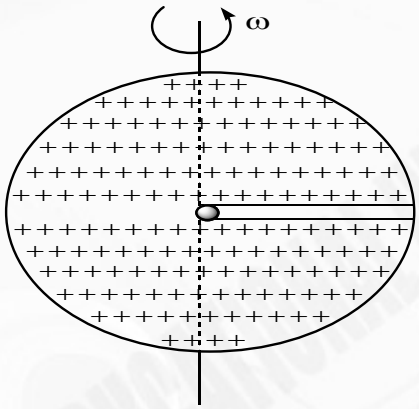


5. A coordinate axis system taking X-axis along horizontal smooth floor is shown in figure. Two small balls of masses m and $3m$ attached with a string are released from some heights on Y-axis as shown in figure. The balls may collide head on or obliquely. After a certain time, mass m is at (9 cm, 20 cm) while mass $3m$ is 25 cm above the X-axis and the string is taut. The balls always remain in XY-plane. If the length of string in cm is $x^2 - x + 1$, then positive value of x is…….



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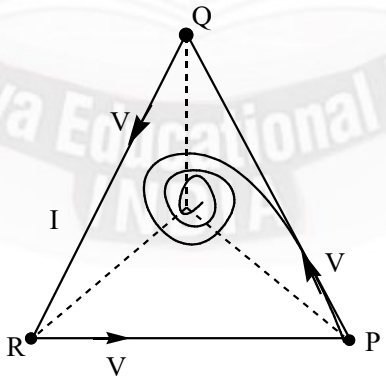
6. A small ball of mass m and charge q is placed at the centre of a uniformly charged Non-Conducting sphere of radius R and charge Q. A narrow smooth groove is made in the sphere from surface to centre as shown in figure. The sphere is made to rotate about a diameter perpendicular to the groove at a constant angular speed ω. Speed w.r.t. ground with which the ball slides out from the groove is (Given that $\frac{KQq}{R} = m\omega^2 R^2$) $\sqrt{p}\ \mathrm{R}\omega$, where p=



SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

7. Three particles P,Q and R initially located at vertices of an equilateral triangle of side start l moving with constant speeds v each such that P always chases (moves towards) Q,Q always chases R and R always chases P. Trajectory followed by P is shown in the figure.
Choose the correct alternative(s).



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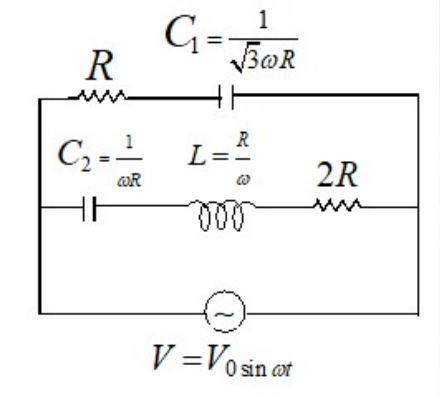
A) Average velocity of P for its motion till the instant it catches Q is $\frac{\sqrt{3}\,v}{2}$

B) Average speed of P for its motion till it completes one revolution around centroid O is v.

C) Distance covered by P when it completes one revolution around centroid O of the triangle is $\frac{2l}{3}\left(1-e^{-2\sqrt{3}\pi}\right)$

D) Radius of curvature of the path followed by any particle at t=0 is $\frac{2l}{\sqrt{3}}$

8. Consider the AC circuit shown and choose correct alternative(s).



A) Effective reactance of the circuit is $\frac{R}{\sqrt{3}}$.

B) Effective impedance of the circuit is $\frac{2R}{\sqrt{3}}$.

C) Current supplied by AC source as a function of time is $\frac{\sqrt{3}V_0}{2R}.\sin\left(\omega t+\frac{\pi}{6}\right)$.

D) Effective wattless current flowing in the circuit is $\sqrt{\frac{3}{32}}\frac{V_0}{R}$.

9. Electric field in the space (xyz) is given by $\vec{E}=-x\hat{i}-y\hat{j}$, find the correct statements.

A) The electric lines are directed towards the origin

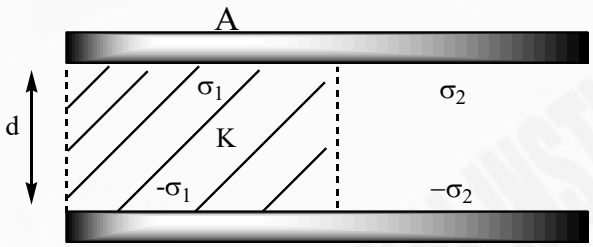
B) If origin is at zero potential then all the points will have -ve potential

C) If a particle of mass m and charge Q is moving on a circle $x^2+y^2=A^2$ then the speed of mass will be $\left(\frac{QA^2}{m}\right)^{1/2}$.

D) The shape of equipotential surfaces will be cylindrical.

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10. A parallel plate capacitor of area A and separation d is charged to potential difference V and removed from the charging source. A dielectric slab of constant K=2, thickness d and area $\frac{A}{2}$ is inserted, as shown in the figure. Let σ_1 be free charge density at the conductor-dielectric surface and σ_2 be the charge density at the conductor-vacuum surface.



A) The electric field have the same value inside the dielectric as in the free space between the plates

B) The ratio $\frac{\sigma_1}{\sigma_2}$ is equal to $\frac{2}{1}$

C) The new capacitance is $\frac{3\varepsilon_0 A}{2d}$

D) The new potential difference is $\frac{2}{3}$V

11. A closed vessel contains a mixture of two diatomic gases A and B. Molar mass of A is 16 times that of B and mass of gas A, contained in the vessel is 2 times that of B:

A) average kinetic energy per molecule of gas A is equal to that of gas B

B) root mean square value of translational velocity of gas B is four times that of A

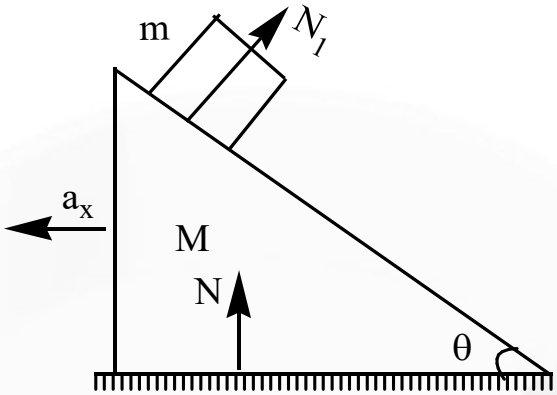
C) pressure exerted by gas B is eight times of that exerted by gas A

D) number of molecules of gas B in the cylinder is eight times that of gas A

12. A block of mass m is released from rest on a wedge of mass M as shown in the figure a_x is the horizontal acceleration of wedge and N is the ground reaction. N_1 is normal reaction between block and wedge.

Mark the correct statements.

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A) $N = (M + m)g$

B) $g - \frac{N}{(M + m)} = a_{cm}$

C) $\frac{N_1 \sin\theta}{M} = a_x$

D) $N_1 = mg\cos\theta$

SECTION – III
(NUMERICAL VALUE TYPE)

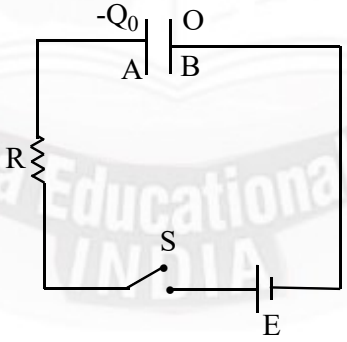
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual

Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

- Answer to each question will be evaluated according to the following marking scheme:

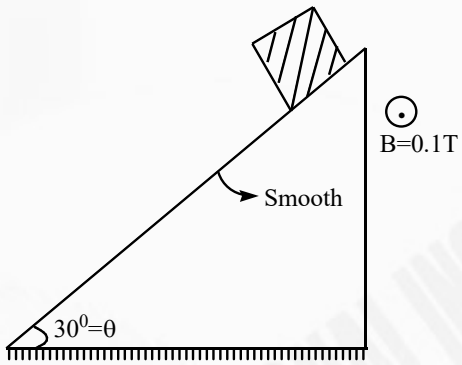
Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 **In** all other cases..

13. The figure shows *RC* circuit with a parallel plate capacitor. Before switching ON the circuit, plate *A* of the capacitor has a charge $-Q_0$ while plate *B* has no net charge. Now, at $t = 0$, the circuit is switched ON. Time at which net charge on plate *A* becomes $\frac{1}{4}$th of intial charge is $RC\ln x$. If $Q_0 = CE$, then x=

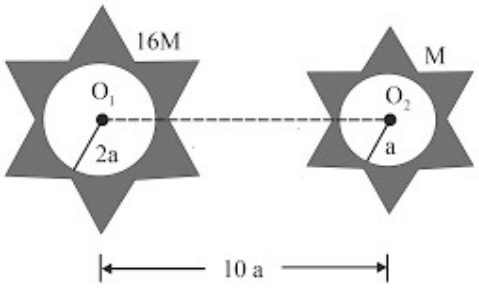


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14. A charged block having specific charge 1 C/kg is released at rest on an smooth long incline in a region with uniform magnetic field. Time (in s) at which the block lose contact with the incline _____

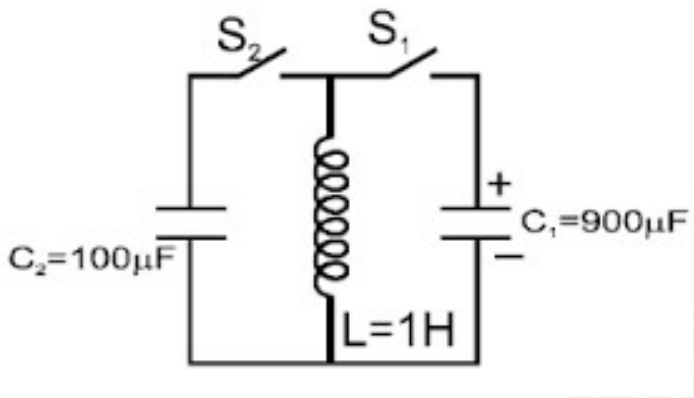


15. Distance between the centres of two stars is *10a*. The masses of these stars are *M* and *16M* and their radii are *a* and *2a*, respectively. A body of mass *m* is fired straight from the surface of the larger star towards the smaller star. The minimum initial kinetic energy for the body to reach the surface of smaller star is $\frac{45GMm}{ka}$. The value of k.



16. In the *L-C* circuit, the capacitor C_1 is charged to a potential 1V and the capacitor C_2 and the inductor were initially energyless. Now, switch S_1 is closed at $t = 0$ and at $t=\frac{1.5\pi}{100}s$, switch S_2 is closed and S_1 is opened. After $\frac{0.5\pi}{100}s$ from this instant S_2 is also opened. Find final potential difference across the capacitor C_2 in volt.

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17. A star initially has 10^{40} deuterons. It produces energy via the processes $_1H^2 + {}_1H^2 \rightarrow {}_1H^3 + p$, and $_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + n$. If the average power radiated by the star is 10^{16} W, in what time (in tera-second) the deuteron supply of the star is exhausted? The masses of the nuclei are as follows:

$M(H^2) = 2.014 amu;\ M(p) = 1.007\ amu;\ M(n) = 1.008\ amu;\ M(He^4) = 4.001 amu.$

18. Let neutron be a point mass and hydrogen atom a solid sphere. A neutron makes a head on collision with a hydrogen atom in ground state kept at rest. The coefficient of for restitution collision is e = 1/2. Find the minimum kinetic energy of colliding neutron (in eV) so that hydrogen atom is excited to higher energy level such that magnitude of electrostatic potential energy in the excited state is one-eighth of K.E. of electron in ground state (mass of neutron = mass of hydrogen).

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CHEMISTRY **Max Marks: 66**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
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Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

19. The ionization energies ($I.E_1$ to $I.E_5$) of 'S' and / or 'P'-block elements (X,Y & Z) are given below

Elements	$I.E_1$ (KJ / mole)	$I.E_2$ (KJ / mole)	$I.E_3$ (KJ / mole)	$I.E_4$ (KJ / mole)	$I.E_5$ (KJ / mole)
X	1086	2353	4620	6223	37830
Y	800	2427	3060	25030	32830
Z	496	4562	6910	9543	13350

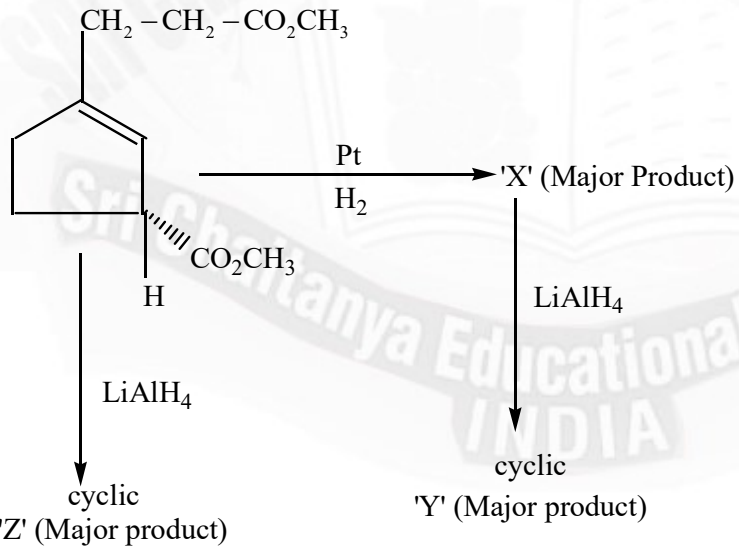
The sum of valence shell electrons in X, Y & Z is

20. The total number of non-polar molecules in which nodal plane in pi bond present only in the molecular plane is

(1) C_2H_2 (2) C_2H_4 (3) $C_2H_2Cl_2 trans$ (4) $B_3N_3H_6$ (5) $C_2H_2Cl_2$ cis (6) C_4H_4 (Buta-1,2,3 triene)

(7) H_2CO_3 (8) C_6H_6 (Benzene) (9) $C_3H_2Cl_2$ (1,3 – dichloropropa 1,2 – diene)

21. Consider the following reaction sequence

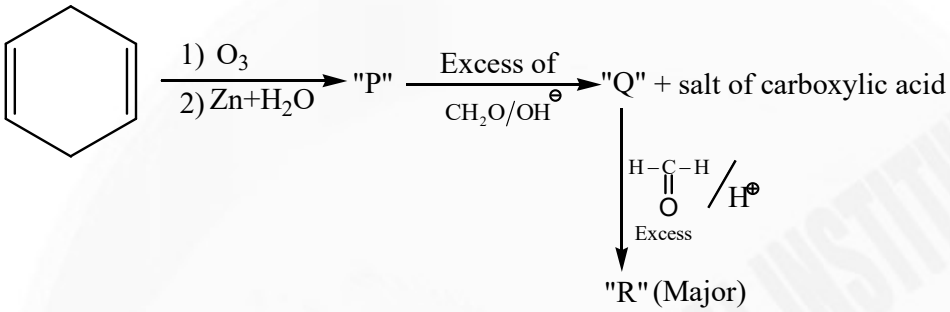


Then the difference in the degree of unsaturation of Z and Y is (Z-Y)

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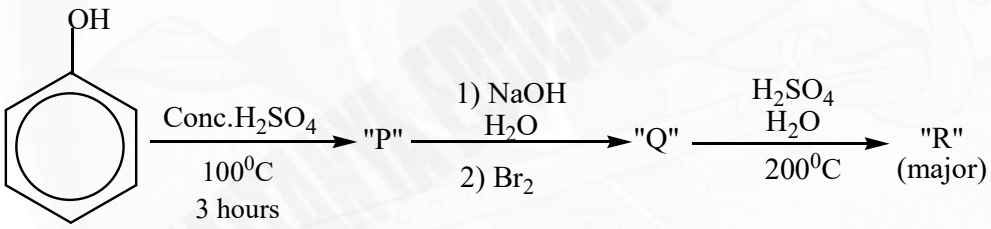
22. In the chemical reaction between stoichiometric quantities of $KMnO_4$ and HCl. what is the number of moles of HCl required for 1 moles of $KMnO_4$ consumed?

23.



Then the number of Sp^3 hybridised carbon atoms in "R" is

24.



Then the number of heteroatoms present in "R" is

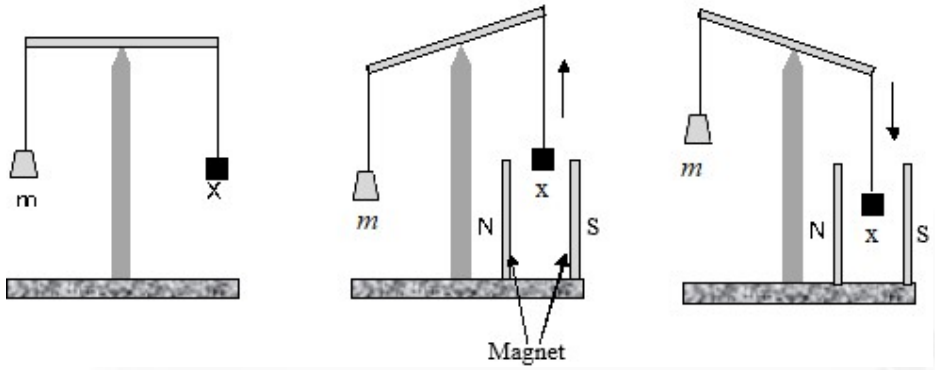
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

25. In an experiment, m grams of a compound **X** (gas/liquid/solid) taken in a container is loaded in a balance as shown in figure **I** below. In the presence of a magnetic field, the pan with **X** is either deflected upwards (figure **II**), or deflected downwards (figure **III**), depending on the compound **X**. Identify the correct statement(s)

(I)	(II)	(III)
Balanced ; Magnetic field absent	Upward deflection ; Magnetic field present	Downward deflection Magnetic field present

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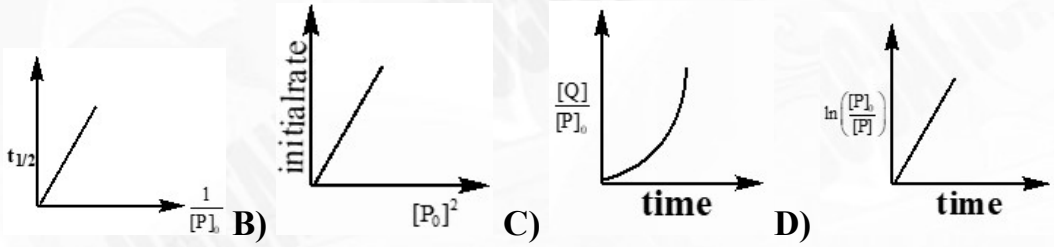
A) If **X** is $H_2O_2(l)$, deflection of the pan is upwards.

B) If **X** is $K_3\left[Fe(CN)_6\right](s)$, deflection of the pan is upwards.

C) If **X** is O_2 (g), deflection of the pan is downwards.

D) If **X** is B_2 (g), deflection of the pan is downwards.

26. Which of the following plots is (are) correct for the given single step reaction 2P → Q
$\left([P]_0 \text{ is initial concentration of P}\right)$



27. Which among the following statement(s) is (are) true for the extraction of aluminum from bauxite?

A) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.

B) Addition of Na_3AlF_6 lowers the melting point of alumina.

C) CO_2 is evolved at the anode during electrolysis.

D) In electro refining of aluminum the layer of pure Al act as cathode.

28. Choose the correct statement(s) among the following.

A) $SnCl_2.2H_2O$ is a reducing agent.

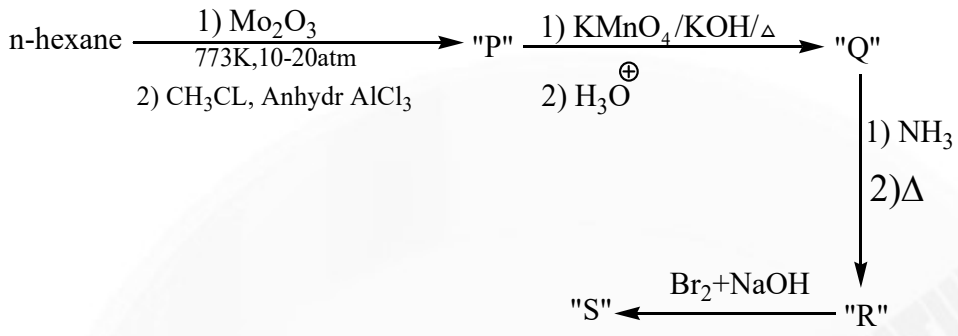
B) SnO_2 reacts with KOH to form $K_2\left[Sn(OH)_6\right]$.

C) SnI_4 orange colour due to LMCT

D) The brown colour precipitate of SnS soluble in yellow ammonium sulphide.

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NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
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29.



Then the correct statement(s) is/are

A) "Q" reacts with ethanol in the presence of $Conc.H_2SO_4$ and produce compound of sweet smell

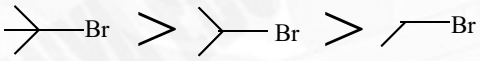
B) "S" reacts with Br_2 / H_2O and gives compound of white precipitate

C) "R" reacts with $LiAlH_4$ gives compound "S"

D) "S" reacts with $NaNO_2 + HCl$ gives "P"

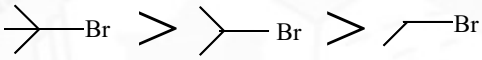
30. The correct order is

A) Order of reactivity of E_1 is as:

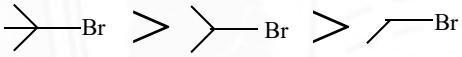


B) Order of nuclephilicity $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$ (InH_2O)

C) Order of reactivity of SN^2 is as:



D) Order of reactivity of SN^1 is as:



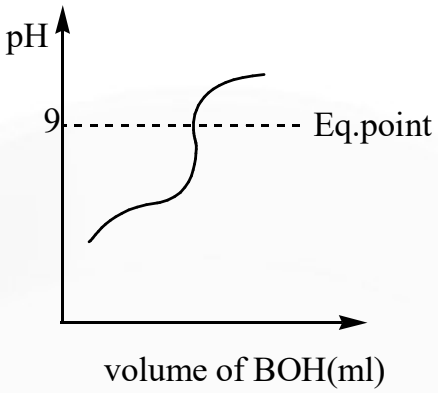
SECTION – III
(NUMERICAL VALUE TYPE)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

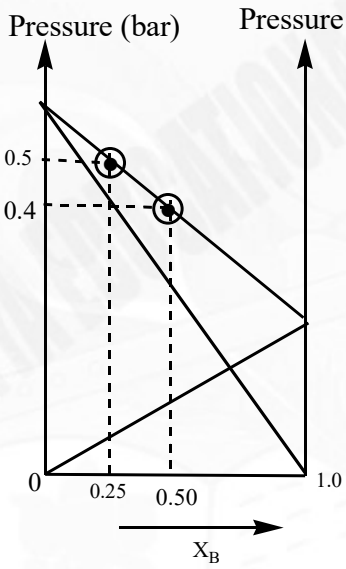
Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 **In** all other cases..

31. A solution of 0.2M weak acid (HA) is titrated with 0.2M of a strong base (BOH). The variation of pH of the solution with the volume of BOH added is shown in the figure below. What is the pK_b of the base (A^-)?

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32. For a liquid solution of A & B following observation is made_____

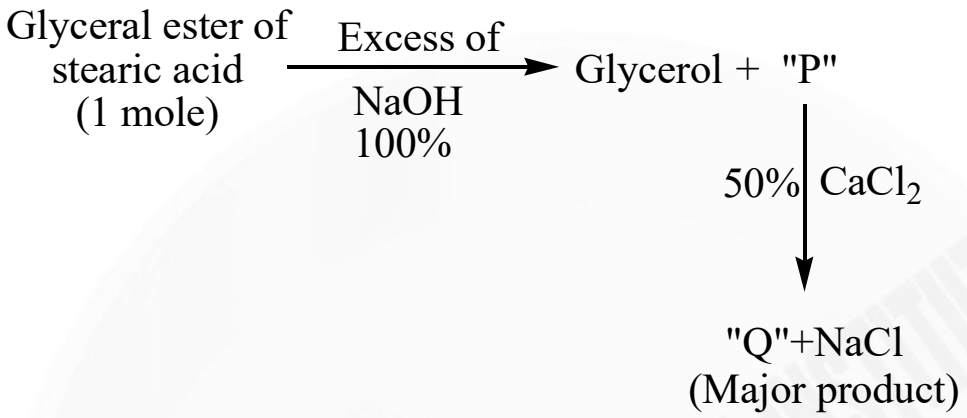


At what pressure (in bar) "$y_A - x_A$" has maximum value [where x_A & y_A are mol fraction of A in liquid & vapour at equilibrium respectively]

33. A photon of 193.42eV strikes with the electron in ground state of ${}^{A}_{Z}X^{(z-1)+}$ & transfers its complete energy to electron due to which electron is excited to higher energy level with its de-broglie wavelength of $2.49\,\overset{0}{A}$. Z is_____

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34.



(Molar mass of C=12.0, H=1.0, O=16.0, Na=23, Ca=40gmol^{-1})

What is the amount "Q" obtained in the reaction is?

35. Zinc can be obtained from Zincite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of Zincite by coke would take place. At 298K: $\Delta_f H^0(ZnO(s)) = -350.0\ kJmol^{-1}$, $\Delta_f H^0(CO(g)) = -110.0\ kJmol^{-1}$, $S^0(ZnO(s))=44\ JK^{-1}mol^{-1}$, $S^0(Zn(s))=42\ JK^{-1}mol^{-1}$, $S^0(C(s))=6.0\ JK^{-1}mol^{-1}$, $S^0(CO(g))=200.0\ J\ K^{-1}mol^{-1}$

Assume that the enthalpies and the entropies are temperature independent.

36. At pH=x, 10^{-3}mol of $Al(OH)_3$ goes completely to $1l$ solution as $Al(OH)_4^-$ while at pH=y, 10^{-3}mol $Al(OH)_3$ goes completely to $1l$ solution as Al^{+3}. x-y =……..

Given $(K_{sp})_{Al(OH)_3} = 10^{-33} M^4$, $(K_{Instab})_{Al(OH)_4^-} = 10^{-34}\ M^4$

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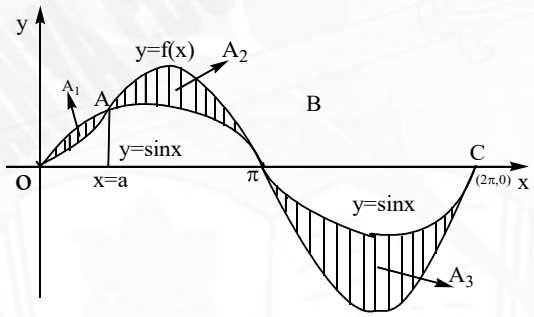
SECTION-I
(INTEGER ANSWER TYPE)

- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks : +3 If ONLY the correct integer is entered;
Zero Marks : 0 If the question is unanswered;
Negative Marks : -1 In all other cases

37. A straight line is drawn through the origin and parallel to the tangent to a curve $\frac{x+\sqrt{a^2-y^2}}{a}=\ln\left(\frac{a+\sqrt{a^2-y^2}}{y}\right)$ at an arbitrary point M. Let the locus of the point P of intersection of the straight line through the origin and the straight line parallel to the x-axis and passing through the point M is S=0, then the area of the director circle of S = 0 is k times πa^2. Let n be the number of digits in k^{2025}, then find the sum of the digits of n.

38. In the adjacent figure, graphs of two functions $y=f(x)$ and $y=\sin x$ are given:



Let $y=\sin x$ intersects $y=f(x)$ at $A(a,f(a))$; $B(\pi,0)$ and $C(2\pi,0)$. $A_i(i=1,2,3)$ is the area bounded by the curves $y=f(x)$ and $y=\sin x$ between $x=0$ and $x=a; i=1$, between $x=a$ and $x=\pi; i=2$, between $x=\pi$ and $x=2\pi; i=3$. If $A_1=1-\sin a+(a-1)\cos a$, then find the value of $(A_3-3A_2+3A_1)$

39. Through a point A on a circle, a chord AP is drawn and on the tangent at A, point T is taken such that AT=AP. If TP produced meet the diameter through A at Q, then the limiting value of AQ when P moves upto A is k times the radius of the circle. Find the number of positive integral factors of k.

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40. Let x_1 and x_2 be the roots of the equation $\sqrt{2025}\cdot x^{\log_{2025}x}=x^2$ and k be the number of zeroes at the end of the product $(x_1\cdot x_2)$. If a continuous function $f:R\to R$ is such that $f(x)=\int_k^x f(t)\,dt$, then find the value of $f(2025^{2025})$

41. Two cards are drawn from a well shuffled pack of 52 cards. If the probability that one of them is a red card and the other is a queen is $\frac{k}{{}^{52}C_2}$, then find the sum of the digits of k.

42. Let $\vec{u}$ be a vector on the rectangular coordinate system with sloping angle $60°$. Suppose that $|\vec{u}-\hat{i}|$ is geometric mean of $|\vec{u}|$ and $|\vec{u}-2\hat{i}|$ where $\hat{i}$ be the unit vector along x-axis, then $|\vec{u}|$ has the value equal to $\sqrt{a}-\sqrt{b}$ where $a,b\in N$, find the value of $(a+b)^2-(a-b)^3$.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

43. Let $f:\left[0,\frac{\pi}{2}\right]\to[0,1]$ be a differentiable function, such that $f(0)=0$ and $f\left(\frac{\pi}{2}\right)=1$ then

A) $f'(\alpha)=\frac{2}{\pi}$ for at least one $\alpha\in\left(0,\frac{\pi}{2}\right)$ **B)** $f(\alpha).f'(\alpha)=\frac{1}{\pi}$ for at least one $\alpha\in\left(0,\frac{\pi}{2}\right)$

C) $f'(\alpha)=\frac{8\alpha}{\pi^2}$ for at least one $\alpha\in\left(0,\frac{\pi}{2}\right)$ **D)** $f'(\alpha)=\sqrt{1-(f(\alpha))^2}$ for at least one $\alpha\in\left(0,\frac{\pi}{2}\right)$

44. Let $f(x)=x^4-8x^3+22x^2-24x$ and $g(x)=\begin{cases}\min f(t):x\le t\le x+1, & -1\le x\le 1\\ x-10 & x>1\end{cases}$, then

A) g(x) is continuous

B) $g(x)\ge -9$

C) g(x) is not differentiable

D) g(x) is non increasing function

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45. There are n straight lines in a plane such that no two of them are parallel and no three of them pass through the same point. Their points of intersection are joined. If the number of maximum **fresh lines** thus introduced is $\frac{n(n-a)(n-b)(n-c)}{d}$ where $a<b<c<d$, then

A) The number of non negative integral solutions of the equation $x_1+x_2+x_3=a+b+c$ is 28

B) The number of non negative integral solutions of the equation $x_1+x_2+x_3=d-b$ is 28

C) The number of ways to express $(abc)^d$ as the product of two positive integers is 41

D) The number of ways to express $(abc)^d$ as the product of two co-prime positive integers is 2

46. Given four points $P_1, P_2, P_3,$ and P_4 on the coordinate plane with origin O which satisfy the condition $\overrightarrow{OP}_{n-1}+\overrightarrow{OP}_{n+1}=\frac{3}{2}\overrightarrow{OP}_n, n=2,3$

A) If P_1, P_2 lie on the curve $xy=1$, then P_3 does not lie on this curve.

B) If P_1, P_2 lie on the curve $xy=1$, then P_3 also lies on this curve.

C) If P_1, P_2 and P_3 lie on the circle $x^2+y^2=1$, then P_4 does not lie on this circle.

D) If P_1, P_2 and P_3 lie on the circle $x^2+y^2=1$, then P_4 also lie on this circle.

47. If $\cot^3\alpha+\cot^2\alpha+\cot\alpha=1$ then

A) $\cos 2\alpha.\tan\alpha=-1$ **B)** $\cos 2\alpha.\tan\alpha=1$

C) $\cos 2\alpha-\tan 2\alpha=1$ **D)** $\cos 2\alpha-\tan 2\alpha=-1$

48. Let p, q, r and s are in A.P and $f(x)=\begin{vmatrix} p+\sin x & q+\sin x & p-r+\sin x \\ q+\sin x & r+\sin x & -1+\sin x \\ r+\sin x & s+\sin x & s-q+\sin x \end{vmatrix}$ such that $\int_0^2 f(x)\,dx=-4$

If d_1 and d_2 be the possible values of the common difference of AP and $g(x)$ be the quadratic equation with d_1 and d_2 as the roots, then

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JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA 300
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A) The range of $g(x)$ is $[0,\infty)$

B) The number of points of non differentiability of the function $|g(x)|$ is 2

C) The number of points of non differentiability of the function $|g(|x|)|$ is 2

D) The number of solutions of $|g(|x|)| = |x|$ is 4

SECTION – III
(NUMERICAL VALUE TYPE)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 **In** all other cases..

49. Let R be the radius of the smallest circular disk large enough to cover every acute isosceles triangle of a given perimeter L, then find the value of R/L

50. If the biquadratic equation $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) has 4 non real roots, two with sum 3+4i and the other two with product 13+i (where $i = \sqrt{-1}$), then find the value of b/100.

51. A tangent to the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P and Q. Find the angle between the tangents at P and Q to the ellipse $x^2 + 2y^2 = 6$ in radians (upto two decimal points)

52. Let the normals at points A, B and C to the parabola $y^2 = 4ax$ are concurrent at the point (h, k). If the equation of circle passing through the points A, B and C is $2(x^2 + y^2) - \lambda(h + 2a)x - ky = 0$, then find $\frac{1}{\lambda}$

53. Circles C_1 and C_2 are externally tangent and they are both internally tangent to the circle C_3. The radii of C_1 and C_2 are 4 and 10 respectively and the centres of the three circles are collinear. A chord of C_3 is also a common internal tangent of C_1 and C_2. Given that the length of the chord is $\frac{m\sqrt{n}}{p}$ where *m, n* and *p* are positive integers, *m* and *p* are relatively prime and *n* is not divisible by the square of any prime, find the value of $(m+n+p)$.

54. If N be the number of natural numbers less than 2025 which can be expressed in the form of $[x\,[x]]$ for some positive real *x*, then find the value of N/100 ([.] denotes greatest integer function).

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31	95	10	98	579

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BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

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